

3. Water treatment in the UK is an important process in making water safe to drink.

- (a) In an experiment to determine the hardness of water in different parts of the country, different technicians tested water samples from Bristol, Birmingham, Manchester and Southampton.

They divided the water samples from each city into two and carried out Test A and Test B on each portion.

Test A

For the **first** portion they followed the method below.

1. Measure 15 cm^3 of water sample into a boiling tube.
2. Add 1 cm^3 of soap solution, insert a stopper then shake vigorously for 5 seconds.
3. Repeat step 2 until lather forms that lasts for 30 seconds. Record the total volume of soap solution added.
4. Repeat steps 1-3 with 15 cm^3 of the other samples of water.

Test B

For the **second** portion they followed the method below.

- Boil the water for 10 minutes.
- Then follow steps 1-4 in **Test A**.

- (i) State the dependent variable in the experiment.

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- (ii) How could the reproducibility of the experiment be checked?

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- (b) The results of the experiment are shown below.

Sample	Volume of soap used to form lather (cm ³)	
	Test A (without boiling)	Test B (after boiling)
Birmingham	1	1
Southampton	11	1
Bristol	12	12
Manchester	1	1

- (i) Explain how the technicians used the results to make conclusions about the type of water in **each city**. [6 QER]

- (ii) Hard water can cause problems in the home.

State **two** problems hard water can cause.

[2]

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2.